

DEPARTMENT OF BUILDING AND ENVIRONMENTAL TECHNOLOGY



RESIDENTIAL BUILDING

**AEBN11 - RESIDENTIAL BUILDING - INTEGRATING THERMAL AND MOISTURE
ASPECTS MEMBERS:**

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Abstract

The study aimed to design and evaluate a multi-story residential building suitable for Sweden's cold climate, ensuring energy efficiency and moisture safety. The study involved the use of advanced simulation tools, including Climate Studio, WUFI PRO 6.8, and Heat 2.0, to analyse and improve building components such as walls, roofs, and foundations.

The methodology included assessing the energy performance, thermal comfort, and hygrothermal behaviour of key building components. Various design alternatives were considered to optimize the placement and orientation of the building, ensuring good solar access while minimizing shading effects on the surrounding context. Passive design measures, such as fixed solar shading and appropriate window-to-wall ratios, were implemented to enhance sustainability.

The results demonstrated compliance with the Swedish building code (BBR) for maximum energy consumption, achieving near-zero energy balance through integrated PV systems. Simulations highlighted effective strategies for maintaining thermal comfort, with indoor temperatures staying within acceptable ranges during summer months. Moisture safety evaluations confirmed the suitability of selected materials and layer configurations, with thermal bridges minimized through detailed Heat 2.0 analysis.

In conclusion, the study provided a comprehensive approach to designing a sustainable residential building that balances energy efficiency, thermal comfort, and moisture safety. The findings underscore the importance of integrating advanced simulation tools into the design process to achieve optimal building performance in cold climates.

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1. Introduction

1.1 Background

Sustainable development nowadays is becoming more and more significant as a global consideration due to climate change and environmental protection. Reducing primary energy consumption has been one of the key objectives in current development efforts.

Buildings account for approximately one-third of the global primary energy supply (Hong et al., 2000), making the reduction of building energy consumption a critical strategy for achieving sustainable development.

To reduce energy input in buildings and minimize greenhouse gas emissions, a concept known as Zero-Energy Building (ZEB) has been adopted as a target by many countries. The European Union is implementing a plan to optimize all new buildings in the region to achieve nearly ZEB standards by 2020 (*Nearly-zero Energy and Zero-emission Buildings*, n.d.).

ZEB achieves net-zero energy consumption by reducing energy usage and compensating for consumption through the generation of energy from sustainable sources. Such buildings make significant contributions to both economic savings and energy cost reduction.

1.2 Objective

The Aim of this report is to analysis a multifamily residential building model located in Lund. Select the suitable site and concept of the building, then simulate the model, output including energy consumption, thermal comfort, moisture safety, thermal bridge and air quality. Figure out the parameters that affect those outputs and utilize some improvements to make the building become ZEB.

2. Climate Study & Site Analysis

2.1 Climate Study

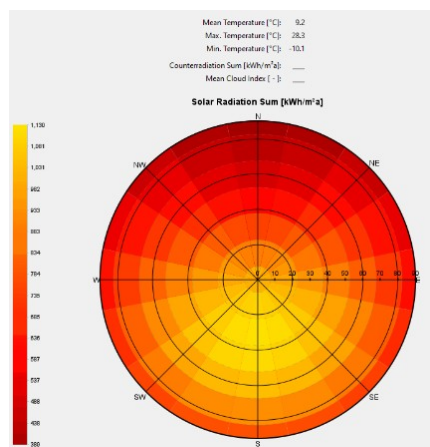
(a) Location

The Brunnshög area in Lund, Sweden is a rapidly developing urban site envisioned for modern residential and commercial projects. The current project focuses on a Multi-Store Residential Apartment, positioned within a mixed-use neighborhood comprising residences, offices, and gardens.

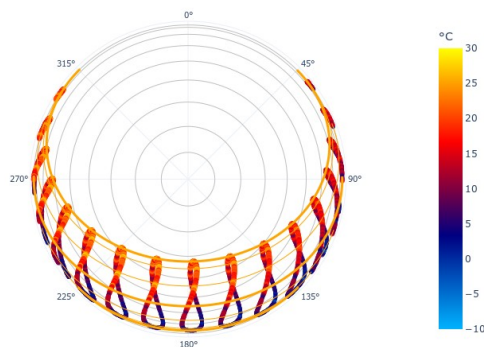
The site is notable for its excellent accessibility, supported by multiple public transportation options, including trams, buses, and roads, making it highly connected to Lund's urban fabric. Geographically, the terrain is flat and barren, with scattered grass and weeds, indicative of its previous use as fertile agricultural land characterized by rich clay and loam soil content.

The Brunnshög district combines strategic location, historical land usage, and future-forward infrastructure, making it an attractive location for sustainable urban growth and development.

(b) Sun Path



The Sun Path Diagram shows the solar altitude and azimuth throughout the year, with peak angles during summer months. The lower solar radiation heatmap reveals values ranging from 0 to 900 Wh/m². Solar intensity peaks between March to September, with maximum exposure during June. Temperatures also rise up to 30°C in summer, while winter sees values as low as -10°C. This analysis aids in optimizing building orientation, daylight utilization, and passive solar energy strategies for year-round performance.



(c) Temperature

Figure.2.1.1 shows the temperature in Lund, and Table2.1.1 lists the highest, lowest and average temperatures in different months throughout the year.

- Dry Bulb temperature:

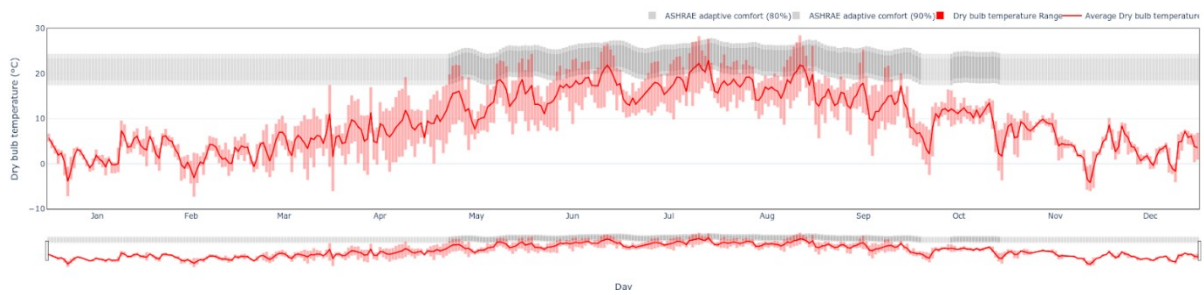


Figure 2.1.1 Monthly temperature in Lund

- Heatmap chart:

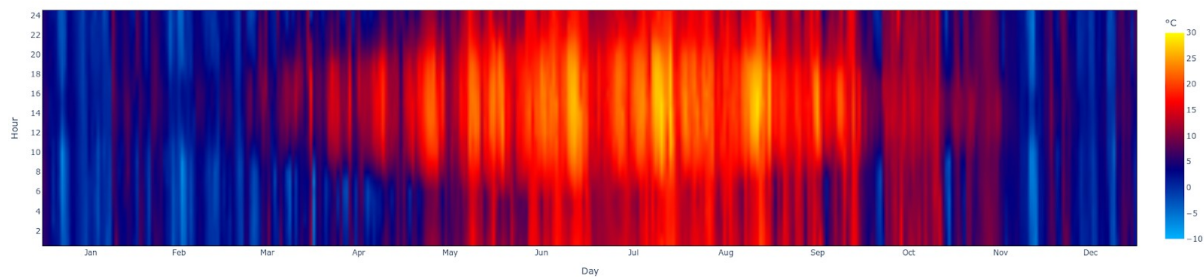


Table 1 The average monthly temperature in Lund

Average	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
High	2°C	2°C	6°C	11°C	16°C	19°C	21°C	20°C	16°C	12°C	7°C	4°C
Temp.	0°C	0°C	2°C	7°C	12°C	15°C	17°C	17°C	13°C	9°C	5°C	2°C
Low	-2°C	-2°C	0°C	3°C	7°C	10°C	13°C	12°C	10°C	6°C	3°C	0°C

In Lund, summers are characterized by cool temperatures, with average temperatures ranging between 15°C and 17°C. The warmest month is July, with an average high of approximately 21°C. Winters, on the other hand, are cold and span from December to mid-March, with average temperatures ranging from 0°C to 2°C.

The climate of this region is influenced by maritime monsoons, resulting in comfortable summer temperatures. However, additional heating is required during the winter months to meet indoor thermal comfort standards.

(d) Average cloud cover

The average cloud cover of Lund is presented in Figure.2.1.2. Overall, Lund experiences relatively high cloud coverage. During the clearest month, July, sky visibility reaches up to 57%. In contrast, the lowest visibility occurs in December, with only 27% of the sky being clear. This indicates that, outside of the summer months, the availability of solar energy for natural lighting and heating in buildings is limited.

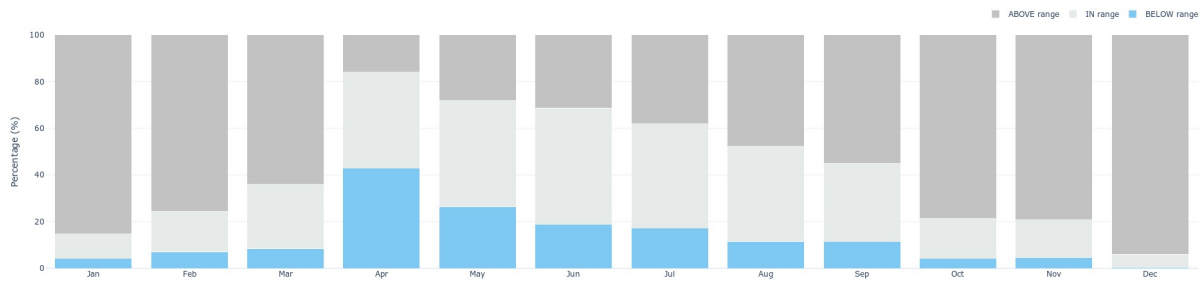


Figure.2.1.2 Average cloud cover in Lund [3]

(e) Humidity and Rainfall

Figure 2.1.4 indicates that humidity in Lund is generally dry for humans, with only 2% of days being muddy in July, while comfortable humidity exceeds 30% from July to August, peaking at 50% in late July. Rainfall remains consistent, averaging 50 mm monthly, with a slight dip to 27 mm between February and March, showing no significant variation overall.

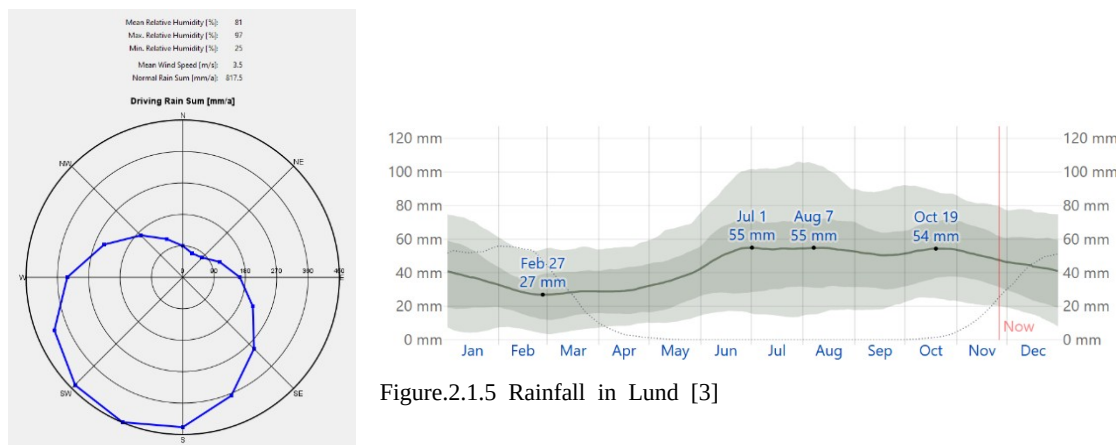


Figure.2.1.5 Rainfall in Lund [3]

(f) Wind Analysis

To investigate the direction and speed of the wind in Lund, Figure.2.1.6 is presented below as reference. The direction of the wind is mostly from west and southwest. In other hand, the average wind speed in Lund stay steady through the year, which is range from 4.6m/s in July to 6.6m/s in January.

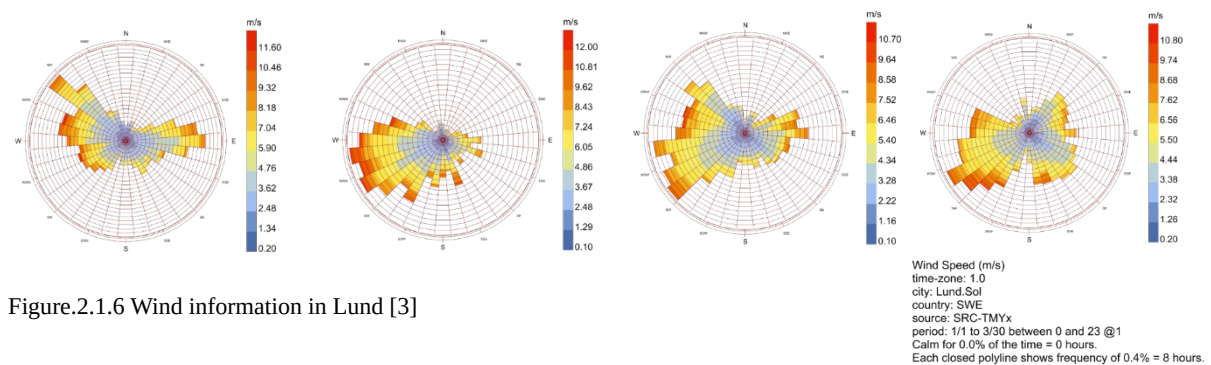
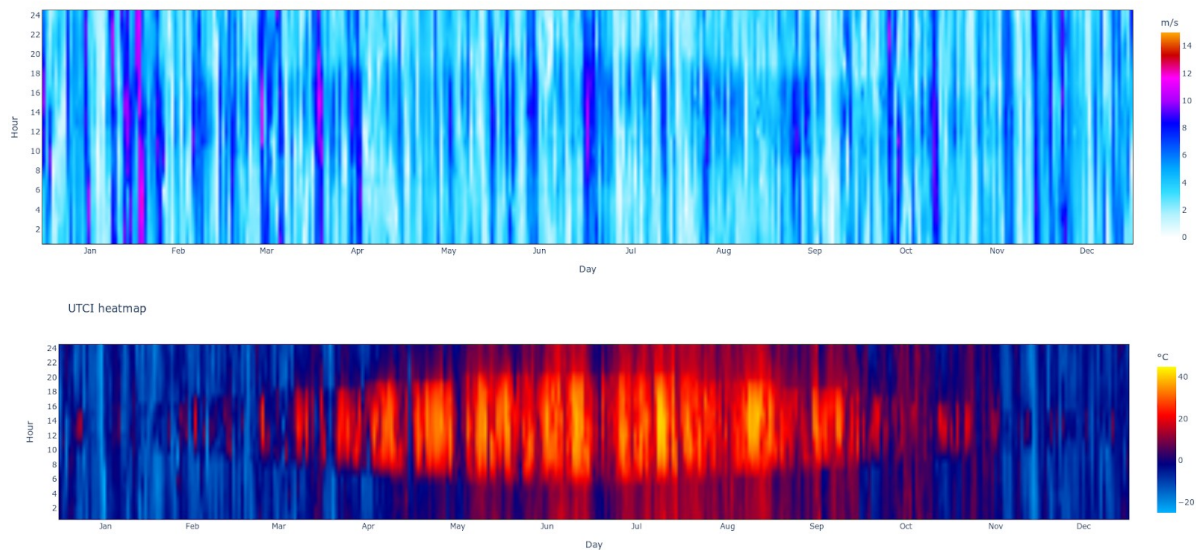


Figure.2.1.6 Wind information in Lund [3]

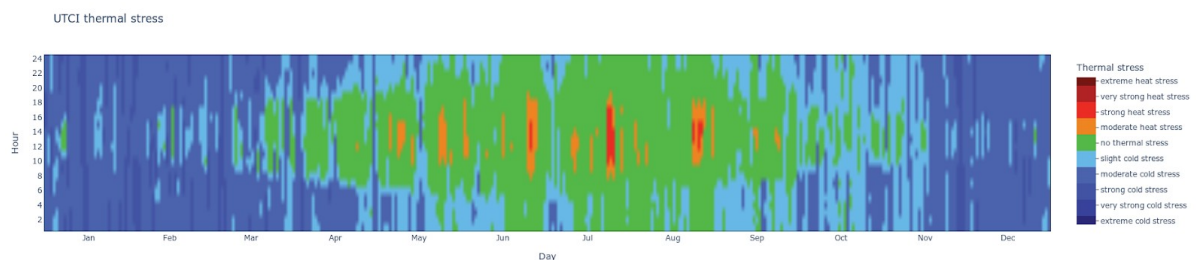
(g) Outdoor Comfort

- UTCI heatmap: Brunnshög, Lund, Sweden



The UTCI (Universal Thermal Climate Index) heatmap highlights thermal comfort throughout the year. Temperatures range from -20°C (deep blue) in winter months like January and December to around 30°C (yellow/red zones) during the summer peak in July and August. The heatmap shows significant thermal comfort variability, with warmer conditions from May to September. This analysis is critical for outdoor space planning, ensuring comfort for residents and visitors year-round.

- UTCI Thermal Stress: Brunnshög, Lund, Sweden



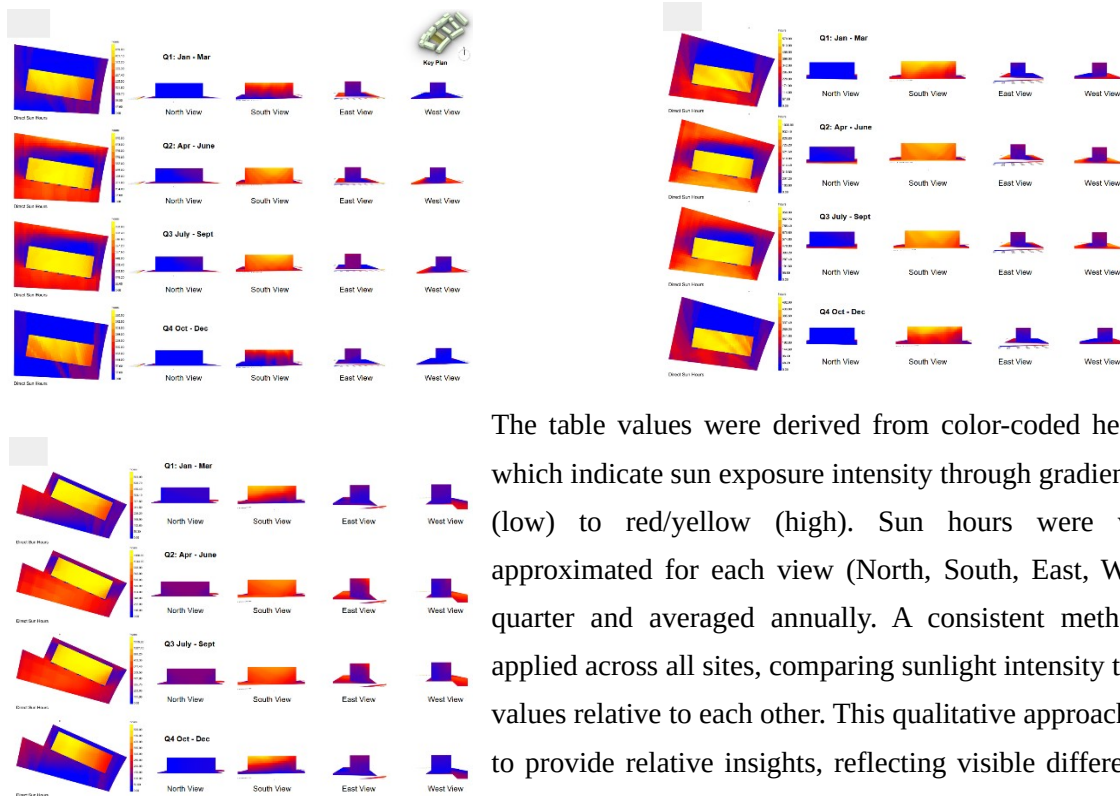
The UTCI thermal stress heatmap illustrates thermal comfort and stress levels throughout the year.

- Winter months (January–March, November–December) show extreme cold stress (dark blue) with temperatures below -10°C.
- Spring to early autumn (April–October) shows no thermal stress (green), particularly in mild months like May and September.
- Peak summer months (July and August) reveal occasional moderate to strong heat stress (orange/red zones).

2.2 Site Analysis

2.2.1 Site Selection

To determine which site would be best for maximizing sun exposure in Lund, Sweden, we need to assess the total sun hours each site receives, along with the distribution across different orientations. Here's a summary of the findings from the compiled data for Sites 1, 2, and 3:



The table values were derived from color-coded heatmaps, which indicate sun exposure intensity through gradients: blue (low) to red/yellow (high). Sun hours were visually approximated for each view (North, South, East, West) by quarter and averaged annually. A consistent method was applied across all sites, comparing sunlight intensity to adjust values relative to each other. This qualitative approach aimed to provide relative insights, reflecting visible differences in exposure between the sites.

To refine the estimates using your specified guidelines:

- **Blue:** Lowest sun hours (assumed around 50 hours or below per quarter).
- **Red:** Moderate sun hours (around 150 hours per quarter).
- **Yellow:** Highest sun hours (around 250 hours or above per quarter).

Using these colour assumptions, I'll recalculate the estimated sun hours for each view direction at each site by quarter and then find the annual average.

Table.2.2.1 Summary data of the different views sun hours

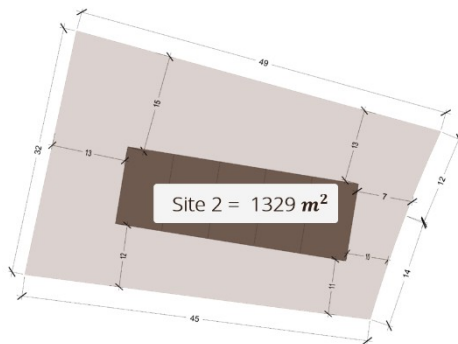
Sun hours

View direction	Site1	Site2	Site3
North	100	100	100
South	200	200	200
East	150	150	150
West	100	150	150

Table 2.2.1 presents updated sun hour estimates for each view direction (North, South, East, and West) across Sites 1, 2, and 3 based on color-coded criteria: Blue (low: ≤ 50 hours/quarter), Red (moderate: ~ 150 hours/quarter), and Yellow (high: ≥ 250 hours/quarter). Key findings include:

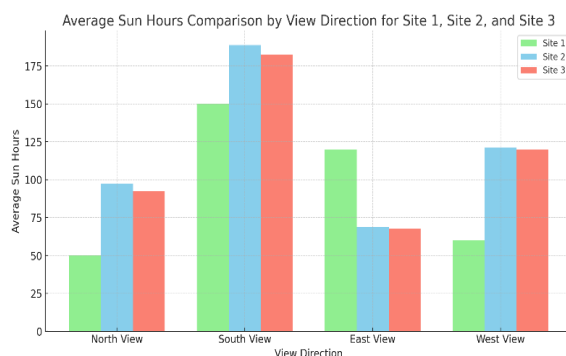
1. **South View Dominance:** All sites receive the most sun on the South view, with Site 2 leading, followed by Site 3 and Site 1.
2. **Overall Sun Hours:**
 - o **North View:** Sites 2 and 3 outperform Site 1.
 - o **East View:** Site 1 receives more sunlight than the others.
 - o **West View:** Sites 2 and 3 have significantly higher exposure than Site 1.
3. **Consistency:** Site 2 demonstrates balanced sun distribution across all orientations, while Site 1 has lower exposure overall.

Recommendation:



In Lund, Sweden's high-latitude climate, maximizing sunlight exposure on south-facing surfaces is crucial for natural lighting and solar energy generation. Based on the analysis:

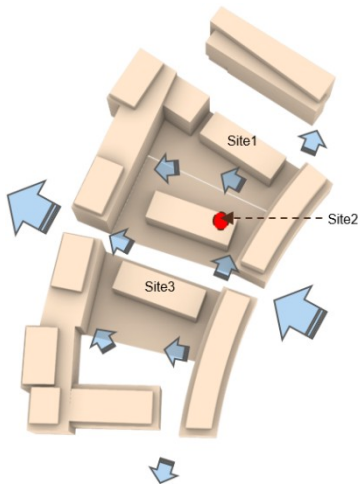
- **Site 2** is the best choice, offering the highest sun hours on the South view, balanced exposure across all directions, and strong performance in low-light winter conditions.
- **Site 3** is a close alternative, with slightly lower sun hours but strong South and West exposure.
- **Site 1** is less favorable due to reduced sun hours, particularly on critical South and West views.



Conclusion:

Site 2 provides optimal sunlight and energy efficiency, particularly beneficial in Lund's variable daylight environment. Bar chart analysis confirms Site 2's superior South-facing exposure, with Sites 2 and 3 outperforming Site 1 in East and West views.

2.2.2 Site and Context Study



The site analysis showcases a strategic layout emphasizing functional connectivity, visual coherence, and sustainable design elements.

- **Centralized Interaction Hub**

The site is organized around a centralized interaction hub, acting as a visual focal point and a space for transition and gathering zones. This design promotes community engagement by encouraging interaction among residents and creating a vibrant social environment. The hub is positioned to ensure accessibility and visibility for all surrounding areas.

- **Spatial Configuration and Height Variation**

The layout incorporates spatial configuration with height variation, enhancing solar exposure and creating dynamic visual interest. Taller buildings are positioned strategically to optimize sunlight penetration to adjacent areas, ensuring equitable natural light distribution. This configuration reduces overshadowing and fosters a harmonious balance between built spaces and open areas.

- **Solar Exposure and Sustainability**

The orientation of buildings maximizes solar exposure, especially for south-facing facades, enhancing energy efficiency through passive heating in colder months. The design integrates shading elements for summer months to prevent overheating, creating a thermally comfortable environment year-round.

- **Movement and Interaction Opportunities**

The design promotes movement and interaction through clear pathways and functional transitions between buildings. The integration of well-defined pedestrian routes enhances connectivity, enabling smooth circulation while fostering an inviting and active urban space.

3. Methodology

3.1 Energy Performance

To obtain energy consumption results that closely reflect reality conditions, the target building was first modeled using Rhino. Basic architectural components such as walls, roofs, and other building elements were created, and multiple zones were defined based on their functional differences.

The next step involved importing the model into Grasshopper for simulation. In this phase, various parameters such as material properties and building orientation were specified. However, utilizing

Grasshopper alone is insufficient to achieve the energy output required. Therefore, the integration of the Ladybug Tools plug-in was crucial.

The input conditions for the model were determined based on the specified requirements and standards. The temperature settings are listed in Table.3.1.1, while the material specifications are provided in Table.3.1.2. It is noticeable that the selection of the U-values for the materials in Table.3.1.2, as well as the detailed material information, will be discussed in subsequent sections.

Table.3.1.1 Temperature setpoint input

Heating set point (°C)	Cooling set point (°C)	COP
21	40	1

Table.3.1.2 U-value setting in model

Envelop	U-value (W/m²K)
Roof	0.089
Wall	0.125
Ground	0.089
Slab	0.087

To determine the heating demand, the internal load of the building needs to be investigated. The following 2 formulas introduce the method that calculates people load and equipment load inside the building.

$$PL = \frac{P \times L}{A}$$

PL = people load (W/m²)

P = people number in the apartment (1.63 for 2-rooms, 2.18 for 3-room)

L = heat load per person (=80W/person)

A = area of the apartment (m²)

$$EL = f \times \frac{E_{equipment} \times 1000}{8760}$$

EL = equipment load (W/m²)

f = usable heat fraction (=70%)

E_{equipment} = household equipment energy use (=30kWh/m²/year)

Except for equipment and people load, energy use for domestic hot water and common lighting and other stuff are also included in the heating demand, however, in this report, they are set to fixed value and turn off in the simulation. Which are 25kWh/m² and 15kWh/m² respectively. The

results of input data of other two load mentioned before are listed in Table.3.1.3.

Table.3.1.3 Energy load input

(Types) Type of Apartment	1	2	3	4	5	Corridor
People Load (W/m ²)	2.16	1.97	1.81	2.11	2.81	0
EL (W/m ²)	2.4	2.4	2.4	2.4	2.4	2.4
Total (W/m ²)	4.56	4.37	4.21	4.51	5.21	2.4

*Types of apartments will be introduced in section4.3

By importing local climate data through an EPW file, the simulation was run to calculate the energy consumption of different building systems, including heating, equipment, and lighting. The simulation outputs included the annual energy use intensity (EUI), providing a comprehensive evaluation of the building's energy performance.

3.2 Thermal Comfort

The thermal comfort temperature ranges from 20°C to 26°C. To investigate the percentage of indoor comfort level, export the operative temperature of different zones in the building model, incorporating data from the EPW file.

These results are typically presented as time series or distribution graphs, illustrating the thermal performance of the spaces over a day or throughout the year. Utilize analytical tools to calculate the percentage of time these results fall within thermal comfort conditions and output the corresponding thermal comfort results.

3.3 Moisture Safety

According to Boverkets Byggregler (BBR) one of the important aspects of building sustainability is ensuring the moisture safety of the thermal shroud. So, another aim of the project was to analyze the moisture safety risks associated with the building envelope and design the construction accordingly. The analysis was assessed by using simulation tools to see how the different parts of the building performed under the existing climate condition in Lund (Boverket, 2023).

3.3.1 Methodology

To conduct the simulation, WUFI was used to analyze moisture conditions for the roof and the exterior wall. Afterwards MATLAB was used to analyze the extract data from WUFI to calculate the risks associated with moisture. Viitanen Model is a method to predict mould growth, also to analyze the moisture and temperature condition in the thermal shroud, such as roof and walls.

3.3.2 Boundary Conditions

In the absence of provided boundary conditions, several assumptions were estimated based on the previous boundary conditions given in the moisture safety course VBFN15. Table 3.3.1 and

3.3.2 illustrate the boundary condition for the roof and the exterior wall.

Table 3.3.1: The details provided present the underlying assumptions for the roof.	
Indoor Climate	The indoor temperature was assumed to be 21°C, with humidity class 2, based on EN 13788 standards
Initial Conditions	WUFI default settings
External Conditions	The exterior SD-value corresponds to Tiles, with no rain adherence and absorptivity values based on wood (spruce) untreaded.
Simulation Time	3 years
Roof Inclination	6°
Roof Orientation	North
Table 3.3.2: The details provided present the underlying assumptions for the exterior wall.	
Indoor Climate	The indoor temperature was assumed to be 21°C, with humidity class 2, based on EN 13788 standards
Initial Conditions	WUFI default settings
External Conditions	The exterior SD-value corresponds
Simulation Time	3 years
Inclination	90°
Orientation	South-west
Building Height	Tall building, middle part up to 10-20 m
Moisture Source	1% fraction of driving rain penetration added to the exterior layer (Wood cladding).

It is important to mention that the most critical layer, both for the roof and wall, was considered be in center the main insulation layer (Cellulose Insulation). This is because the drying process in the middle of the insulation will be much more difficult, due to limited airflow and restricted evaporation paths, if moisture accumulates so long in the main insulation layer. For this reason, the middle of the Insulation was identified as the most critical layer. Therefore, the simulations were focused on assessing the overall moisture impact on the mid of the cellulose insulation for respective elements, the results are illustrated in the coming section (5.3).

3.4 Thermal Bridge

Thermal bridge is another way for heat loss and this type of heat loss is unavoidable but through a proper construction design the percentage could minimize the loss. The most common linear thermal bridge (denoted by the Greek letter Ψ , pronounced "psi," and measured in W/m K) typically occurs at the edge beam, which supports the outer wall. In wooden constructions, significant linear thermal bridges can also arise at the connections between intermediate floors, the roof, and the outer wall.

Additionally, thermal bridges often form between window or door frames and the surrounding wall. This is primarily due to extra framing elements and the three-dimensional heat flow that occurs in window and door recesses (Boverket, 2012).

3.4.1 Methodology

To calculate the thermal bridge a program was used, called HEAT2, which is a two-dimensional transient and steady state heat transfer. With help from HEAT2 the simulations were performed, and the data for heat transport (q -value in W/m^2) were collected to calculate the Ψ -value for wall with roof, intermediate slab and the ground slab.

3.4.2 Boundary Conditions

The boundary conditions in the HEAT were provided according to the guidelines to facilitate the steady state calculation of thermal bridge, Table 3.4.2 shows the boundary conditions for each and every joints in the building envelope.

Table 3.4.2: Table outlines the boundary conditions				
T [$^{\circ}C$] & R [$m^2 \cdot K/W$]	T_{out}	T_{In}	R_{se}	R_{si}
Connection between Wall and Roof	0	21	0.04	0.1
Connection between Wall and Intermediate Floor	0	21	0.04	0.13
Connection between Wall and Ground Slab	9.3	21	0.04	0.17

3.4.3 Experimentation Stage

3.4.3.1 Comparison of different insulations

Insulation is the main part of good thermal conductivity, with this so works different type of insulation differently. In this case, it was already decided that the load-bearing construction system would be made of wood (CLT). Evaluating how various insulation materials performed within the existing system was crucial for ensuring moisture safety. Therefore, several insulations were tested such as, Extruded polystyrene (XPS), Expanded polystyrene (EPS), Mineral Wool and Cellulose insulation (Figure 3.4.3.1).

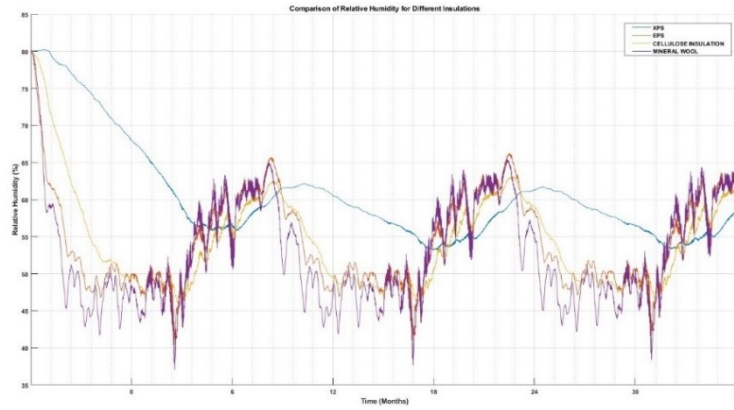


Figure 3.4.3.1.1: A graph presenting RH for different types of insulations for 3 years (Roof).

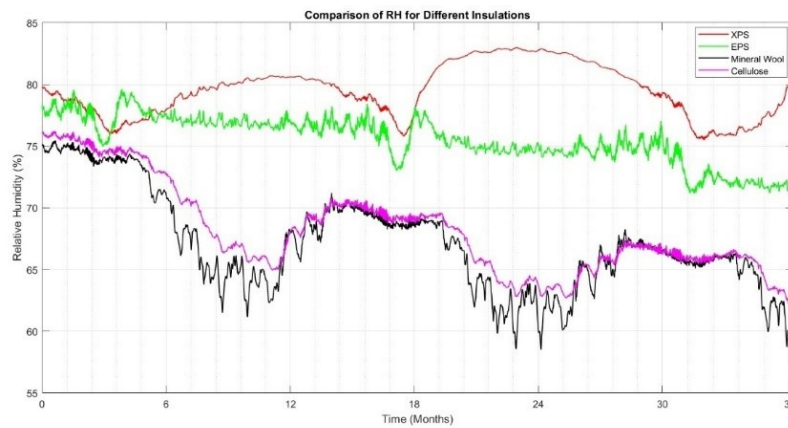


Figure 3.4.3.1.2: A graph presenting RH for different types of insulations for 3 years (Wall).

3.4.3.2 Comparison of different Air Changes per Hour [ACH]

One of the main factors that plays a crucial role in moisture safety in construction is the ventilation efficiency of the air layer. The reason for that is because the air movement helps in drying out the construction faster. Simulations were conducted to observe and compare the effects of different Air Changes per Hour (ACH) values: 10, 20, 30, and 40 ACH (Figure 3.4.3.2).

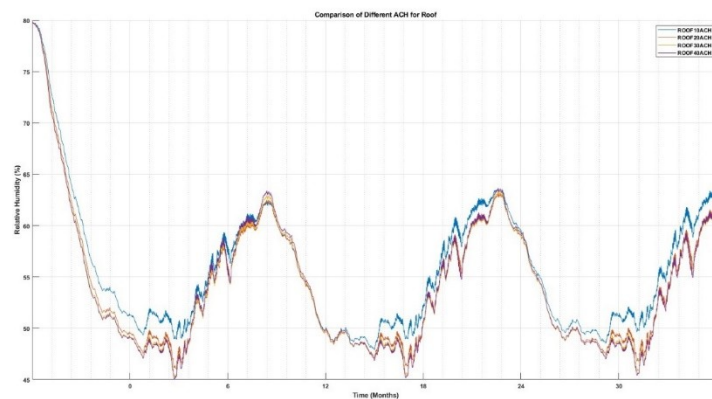


Figure 3.4.3.2.1: A graph presenting the RH for different ACH for 3 years (Roof).

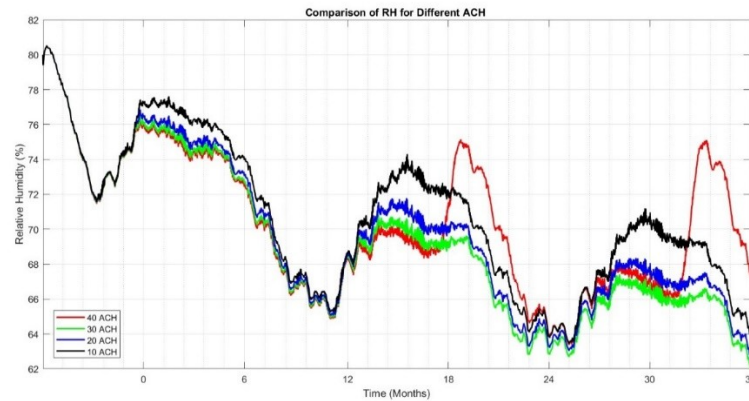


Figure 3.4.3.2.2: A graph presenting the RH for different ACH for 3 years (Wall).

4. Design Development

4.1 Intermediate Design & Output

The perspective view and floor plan of the intermediate design is presented in Figure.4.1.1. The intermediate building includes 4 floors, except for the first and second floor, the area of different floors are diverse.

Moreover, there is only one apartment on the 4th Floor. The PV panels are on the top surface of each floor of the building, which is unreasonable due to the unable of getting to the place. In that case, the final design refines this part.

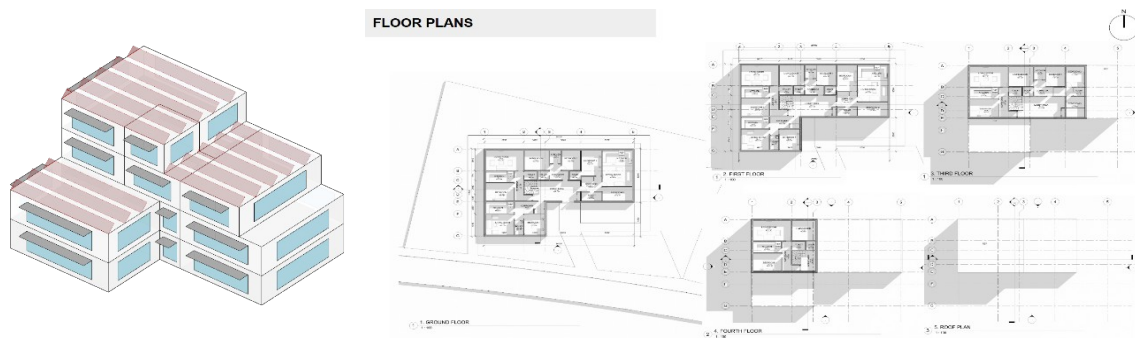


Figure.4.1.1 Perspective view and floor plan of intermediate design building

4.1.1 Energy & Moisture Performance

The performance of initial design is demonstrated in Table.4.1.1 and Table.4.1.2

Table.4.1.1 Energy performance of intermediate design building

	Value (kWh/m ² heated area per year)
Energy Use Intensity (EUI)	33.5
Heating Demand	28.3

Table.4.1.2 Initial wall data

Parameter	Value
Thickness	398mm
U-value	0.15 W/m ² K
RH< RH-critical	80%
Critical layer	Mineral wool layer
Mould index	Steady at 0
Mould growth rate	0 for All Periods
Thermal bridge	0.00186W/mK

*Thermal bridge for wall with slab < 0.4 W/m·K

4.1.2 Critical Considerations

Entrance: The door position, especially on the south side, should be redesigned as a room to reduce heating load and utilize natural heat efficiently for the indoor living room.

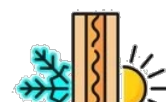
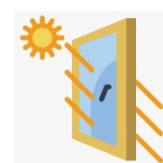
Orientation of Rooms: Room positions should be reorganized as they contribute to the overheating rate, exceeding the required limits.

PV Panel: PV panels on adjacent floors should be reconsidered as they occupy outdoor spaces like balconies and pose maintenance challenges.

4.2 Final Design

4.2.1 Key Concept of Design Form Derivations

1. Focus on South-Facing Surfaces:
 - o Maximizes solar gains during winter for passive heating.
 - o Incorporates shading solutions in summer to prevent overheating, ensuring year-round thermal comfort.
2. Roof Design:
 - o Optimized for solar panels, leveraging long daylight hours in summer to enhance energy efficiency.
3. East/West Facades:
 - o Incorporates shading devices to balance sunlight intensity,



reducing glare and minimizing thermal discomfort.

4. North Facade:

- Prioritizes insulation to compensate for limited sunlight, maintaining indoor thermal stability in colder conditions.

This design approach ensures energy efficiency, thermal comfort, and sustainability while responding to the building's orientation and environmental context.

4.2.2 Design

The final design architectural perspective is illustrated in Figure.4.2.1, while Figure.4.2.2 presents the rendered visualization of the new design. This design considers the impact of lighting, incorporating balconies for natural illumination and corresponding shading for each apartment.

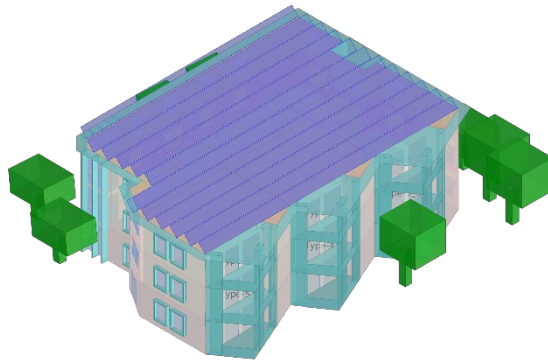
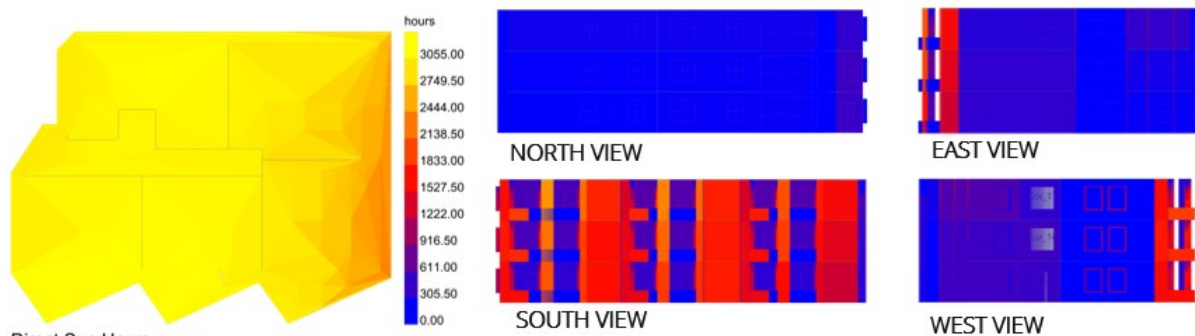


Figure.4.2.1 Perspective view of Final design

Compared to the preliminary design, the new proposal features equal floor areas and an identical number of apartments per level. Additionally, solar panels are centralized on the rooftop of the building, which is a more rational arrangement than in the earlier design.

Another notable difference lies in the positioning of the building entrance. In the preliminary design, the entrance faced south, an advantageous orientation for room layouts. However, this placement resulted in an inefficient use of resources for the hallways. In the new design, the entrance orientation has been revised to address this issue.

4.2.3 Direct Sun Hours Study



Orientati on	Sun Hours (Low)	Sun Hours (Moderate)	Sun Hours (High)	Key Insight
North View	Predominantly Blue (~0-611)	None	None	Minimal sunlight due to orientation

East View	Mostly Blue (~0-611)	Red Bands (~1222-1833)	Minimal	Captures morning sunlight with moderate exposure
South View	Minimal (~0-611)	Dominant Red (~1222-1833)	Dominant Yellow (~3055)	Maximizes sunlight, ideal for solar gains
West View	Mostly Blue (~0-611)	Red Bands (~1222-1833)	Minimal	Captures afternoon sunlight with moderate exposure

This analysis shows optimal solar gains on South-facing surfaces, moderate on East/West, and minimal on the North. These findings guide facade designs and shading strategies for energy efficiency and daylighting.

4.2.4 Typical Floor Plan

The typical floor plan of Final design building is shown in Figure.4.2.3. Each floor consists of five apartments, each with a unique design. Among these, three of the apartments are 2-room type, while the remaining two are 3-room apartments. All apartments are equipped with a bathroom, kitchen, balcony, bedrooms, and a living room.



Figure.4.2.3 Typical floor plan of redesigned building

4.2.5 Design Views



Figure.4.2.2 Final design model in render

4.2.6 Window Selection

Parameter	Values
WWR	35%
U-value	0.57 W/m ² K
SHGC	0.216
Tvis	0.512
Glazing type	Triple
Gas	Air

The Solarban 72(2) Triple Layer (Argon-filled) glass is the optimal choice for Brunnshög's project, maximizing energy efficiency, thermal comfort, and sustainability. However, the standard Triple Layer (SHGC 0.216) glass provides a cost-effective alternative with solid performance.

4.3 Zone Breakout

The zone breakout of the building is shown in Figure.4.3.1. They are divided by each apartment and the corridor. Each floor includes 5 types of apartments and 1 corridor.

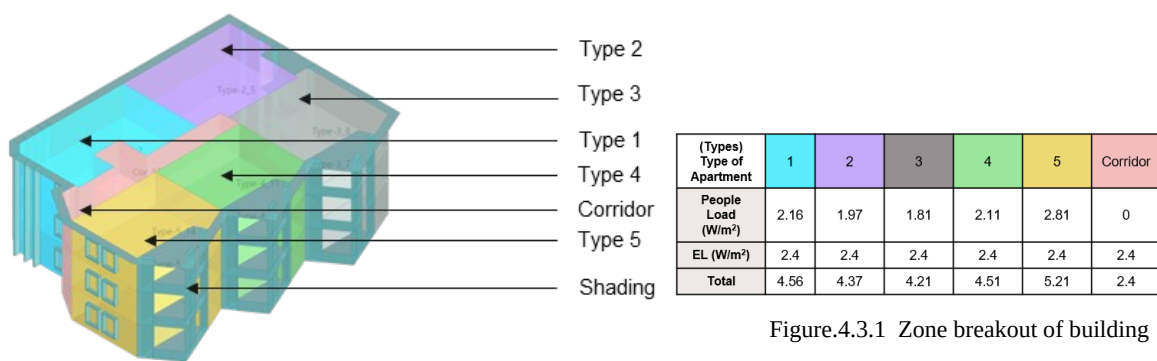


Figure.4.3.1 Zone breakout of building

Figure.4.3.2 presents the typical floor plan of the building, and the details of each apartment are exhibited in Table.4.3.1.



Figure.4.3.2 Typical Floor Plan

- **Shading Integration:**

Shading is integrated into the design to reduce thermal gains, which is crucial for balancing cooling loads during peak summer months.

4.4 Zone Breakout (Floor 1 – Southwest)

To analysis energy and thermal performance in detail, the zone breaking out of the most critical apartment (1F, southwest) is necessary, which is presents in Figure.4.3.3. This apartment is a 64m², type is 2-rooms with 10m² window.

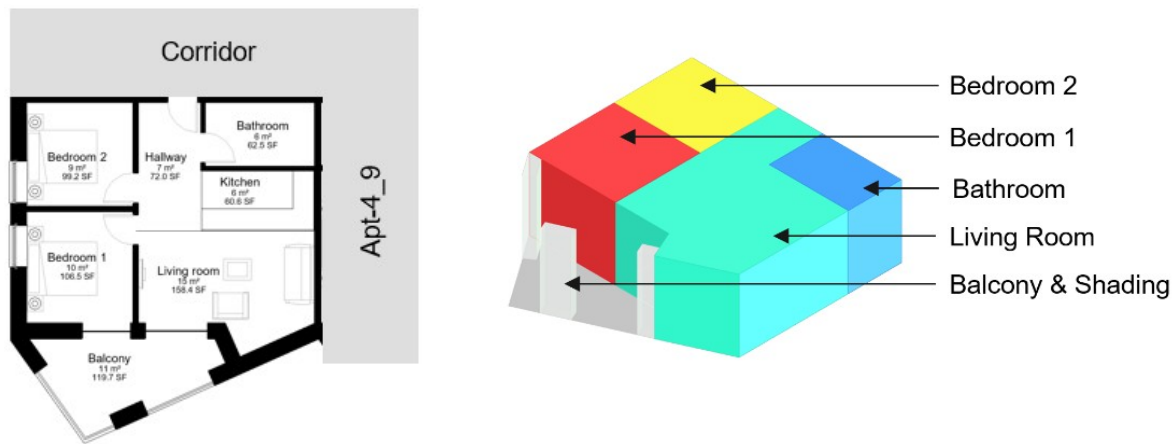


Figure.4.3.3 Floor plan and Zone Breakout of one apartment (1F Southwest)

- FLOOR AREA: **64m²**
- WINDOW AREA: **10m²**

ZONES	AREA (m ²)
Bedroom1	10
Bedroom2	9
Bath Room	6
Living Room	28
Balcony	11

5. Result

5.1 Energy Performance

The output of Heating demand of the Final design building in simulation is presented in Figure.5.1.1. The equipment load intensity is steady at approximately 2.7 kWh/m². While heating load intensity reaches the maximum which is 3.0kWh/m² in January then gradually drops to 0 on June, after staying 3 months with no heating load, the value starts to growth from September to January next year.

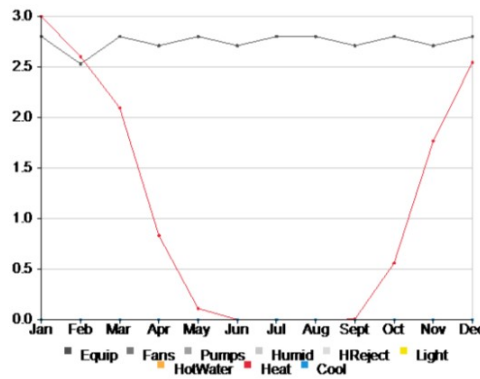


Figure.5.1.1 Heating Demand of Final Model (Monthly)

The simulation results for the heating demand in different zones are presented in Figure.5.1.2, with detailed data listed in Table.5.1.1 and Table.5.1.2. The following observations can be summarized: the apartment with the highest energy consumption is Apartment 3, with a heating demand of 826.0 kWh, while the lowest is Apartment 11, with a heating demand of 116 kWh. The energy consumption in the corridors is generally lower than that in the apartments, ranging from 85.9 kWh to 316.7 kWh.

Table.5.1.1 Zone energy demand (apartments)

APARTMENT															
Zone	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Energy demand (kWh)	682.2	408.6	826	535.9	277.4	679	452.7	238.9	571.2	233.5	116	314.2	455.5	256.6	256.9

Table.5.1.2 Zone energy demand (corridor)

CORRIDOR			
Zone	1	2	3
Energy demand (kWh)	256.9	85.9	316.7

The most critical apartments that required to be analysis independently is the apartment in southwest side in 1F. The output of energy performance of this apartment is presented in Table.5.1.3.

Table.5.1.3 Heating demand

Heating Demand	Heating Energy
11.57 kWh/m ² heated area per year	501.15kWh

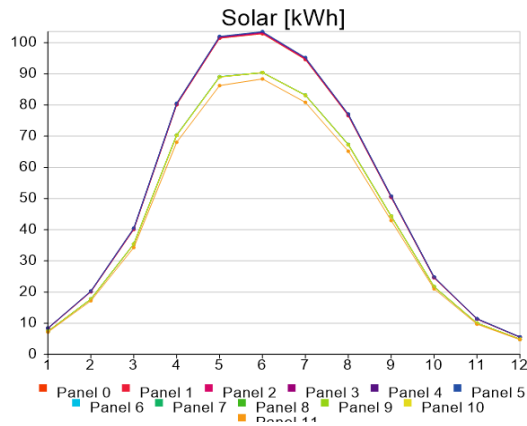


Figure.5.1.3 Energy generated by PV panel

The solar energy generated by PV panels are presented in Figure.5.1.3. It can be investigated that during warm months, there is more energy that produced by the panel, especially May and June, which is over 100kWh. During December to January next year, the solar energy generated by panels is lower than 10kWh. The total area of panels is 294.09 m², and energy generated through the year is 29010kWh. The energy balance is exhibited in Table.5.1.2. It can be read that approximately 88% of energy consumption can be supplied by the building itself.

Table.5.1.2 Energy balance

Parameter	Energy per year (kWh)
Energy produced by PV panel	29010
Building energy consumption	32832
Energy balance	3822

5.2 Thermal Comfort

To evaluate indoor comfort levels, periods with temperatures exceeding 26°C were identified as overheating, and the temperature variations within the model were recorded. Table.5.2.1 presents the overheating rates for different zones, defined as the proportion of time the temperature exceeded 26°C relative to the total duration.

Table.5.1.1 Overheating rate (apartments)

Apartment															
Zone	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Overheating (%)	0.82	12.16	6.51	0.11	4.92	1.66	0.3	7.59	1.73	0	1.59	0.21	0.21	4.1	0.52

Table.5.1.2 Overheating rate (corridor)

Corridor			
Zone	1	2	3
Overheating (%)	0.00	0.00	0.62

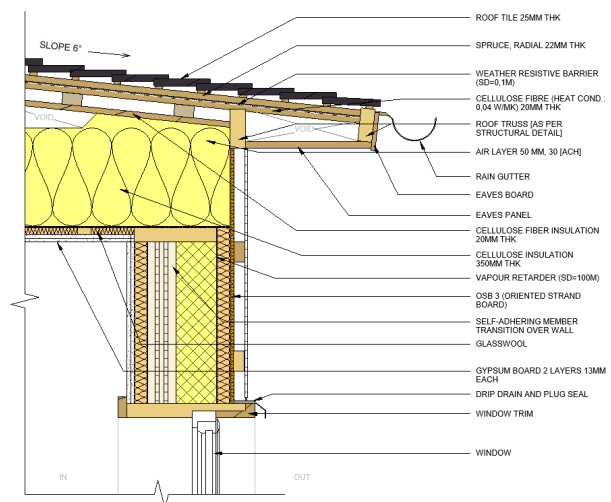
The results indicate that seven apartments, nearly half of the total, have overheating rates below 1%. Notably, Apartment 2 has an overheating rate of 12.16%, significantly higher than the second-highest rate of 7.59% observed in Apartment 8. This rate is well above average and highlights the need for further optimization in Apartment 2 to improve thermal comfort. The overheating rates for corridors on all three floors are below 1%, indicating low levels of solar radiation. Therefore, no special interventions are necessary.

After breaking out a critical apartment (Southwest, 1F) and simulated, the temperature range throughout the year in this apartment is from 20.48°C to 23.79°C, which indicates that there is 0% overheating.

5.3 Moisture Safety Design

5.3.1 Roof Construction Detail

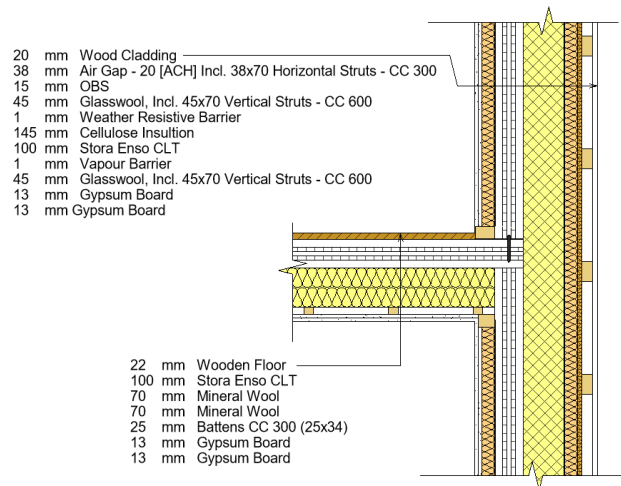
- Roof:
 - Layers of roof tiles, weather-resistant barriers, and cellulose fiber insulation for thermal and moisture control.
 - Spruce wood rafters provide structural support.
- Wall:
 - Layers of gypsum board, oriented strand board (OSB), vapor retarder, and cellulose insulation for thermal performance and structural integrity.
 - Air gaps enhance ventilation.
- Drainage:
 - A rain gutter and drip edge prevent water ingress and direct rain away from the building.



5.3.2 Slab on Wall Detail

- **Wall Components:**

- o **External Layers:** 20 mm wood cladding with a 38 mm ventilated air gap and 10 mm OSB for durability and moisture control.
- o **Insulation:** Multiple layers of glass wool (45 mm) and cellulose insulation, ensuring thermal performance.
- o **Structural Core:** 145 mm Stora Enso CLT (Cross-Laminated Timber) provides strength and stability.
- o **Internal Layers:** Vapor barrier, gypsum board (13 mm), and battens for interior finishing and additional insulation.



- **Floor Components:**

- o Includes a 22 mm wooden floor and layered mineral wool for thermal separation and acoustic dampening.

5.3.3 Ground Detail

- **Floor Layers:**

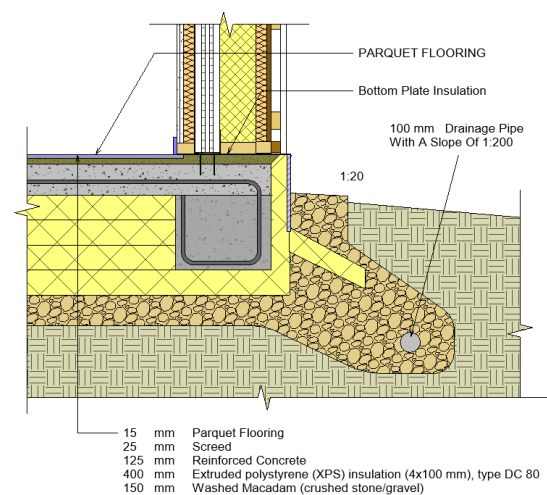
- o **Surface:** 15 mm parquet flooring and 25 mm screed for durability and finish.
- o **Structural Base:** 125 mm reinforced concrete ensures strength and stability.

- **Insulation:**

- o **Extruded Polystyrene (XPS):** 400 mm thick insulation (type DC 80) prevents heat loss through the foundation.

- **Drainage and Moisture Control:**

- o A 100 mm drainage pipe with a 1:20 slope efficiently manages groundwater.



- A 150 mm layer of washed macadam (crushed stone/gravel) improves drainage and prevents water accumulation.

(a) Roof

Figure.6.2.1 presents the Relative humidity (RH) and RH critical of the roof. The RH critical value of the wall remains consistently above 80%. While the actual RH fluctuates between 46% and 64%. During the warmer months (April to September), the RH tends to be slightly lower, stabilizing around 50%, whereas in the colder months, it increases to approximately 60%.

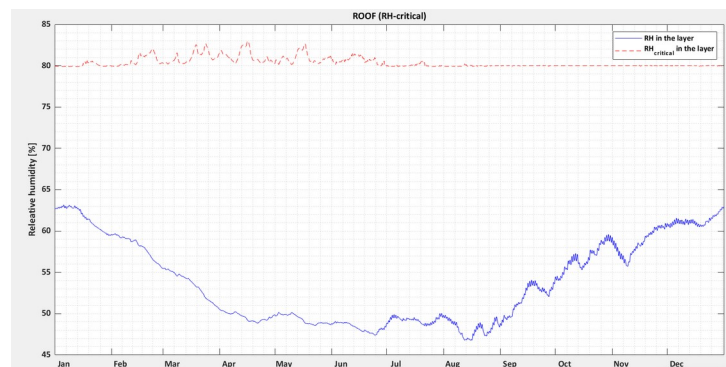


Figure.6.2.1 RH and RH critical of the roof

At all times throughout the year, the RH values remain significantly below the RH critical threshold. This indicates that the roof has no risk of reaching moisture levels that could lead to condensation or mold growth, even during colder months when relative humidity typically rises.

Same results can be concluded from Figure.6.2.2 and Figure.6.2.3, which are the mold index and mould growth rate. Figure.6.2.2 shows that the mould index stays at 0 for whole year, this means that there is no moisture risk. Moreover, Figure.6.2.3 indicates that the growth of moisture will always be 0, in that case, there is no detectable growth of mold or other moisture-related issues in the roof.

Figure.6.2.5 shows the mould index of the wall, it can be read that the value of that also be 0, meaning that there is no risk of mould. Figure.6.2.6 exhibits the moisture growth rate during a year, the rate is constant in all of the days throughout the year, in that case, there is no moisture increase inside the wall.

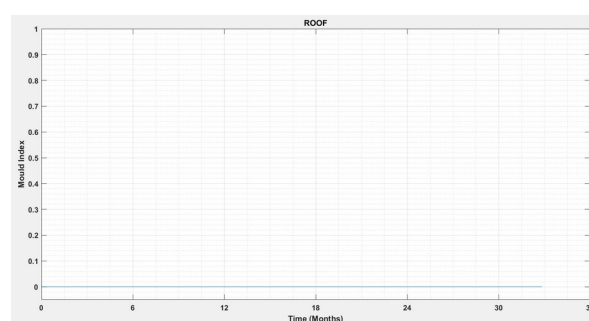


Figure.6.2.2 Mould index of the roof

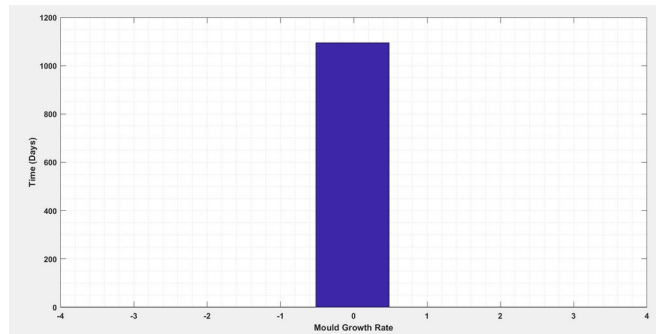


Figure.6.2.3 Moisture growth rate in the roof

(b) Wall

Similar to the roof, the RH critical of the walls also remains consistently above 80% in Figure.6.2.4. However, unlike the roof, the RH critical for walls is slightly higher during the colder months (October to April), with peak values reaching approximately 85%. The actual RH fluctuates between 61% and 66%, with average values during the warmer months (June to November) slightly higher than those in other months.

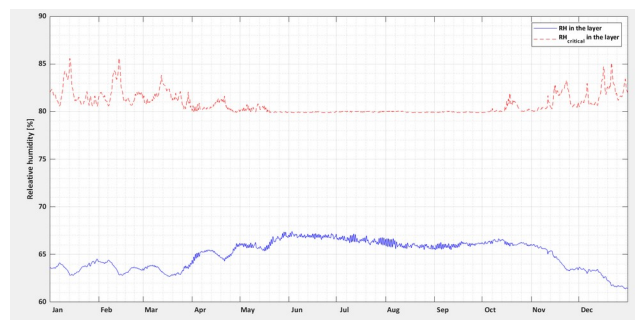


Figure.6.2.4 RH and RH critical of the wall

The RH values consistently remain well below the RH critical threshold, indicating excellent waterproofing performance of the wall system, with no potential humidity risks. Seasonal environmental variations do not have any significant impact on the wall's performance, further emphasizing its robust hygrothermal properties.

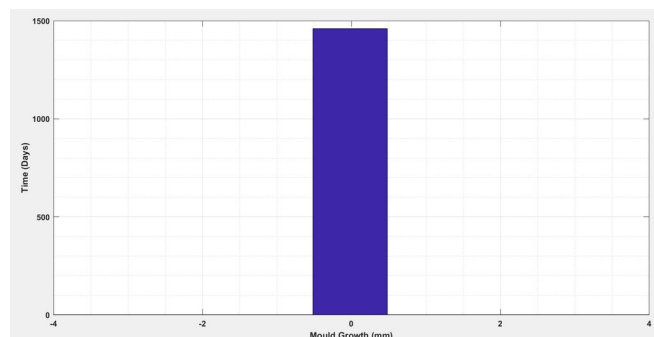


Figure.6.2.5 Mould growth of Wall

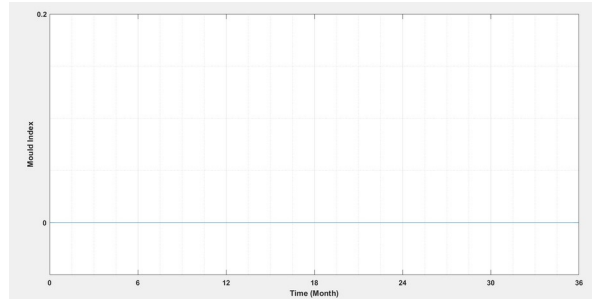


Figure.6.2.6 Mould index of wall

5.4 Thermal Bridge

In the analysis of the building envelope, thermal heat loss was evaluated some critical junctions like the connections between the wall and roof, the wall and intermediate slab and the wall and the ground slab. The following sections provide an overview of the simulations and results of the calculations:

5.4.1 Thermal Bridge Calculation for Connection Between Wall and Roof

In this study the thermal bridge analysis focused on one of the three key junctions, as it is illustrated in Tabi

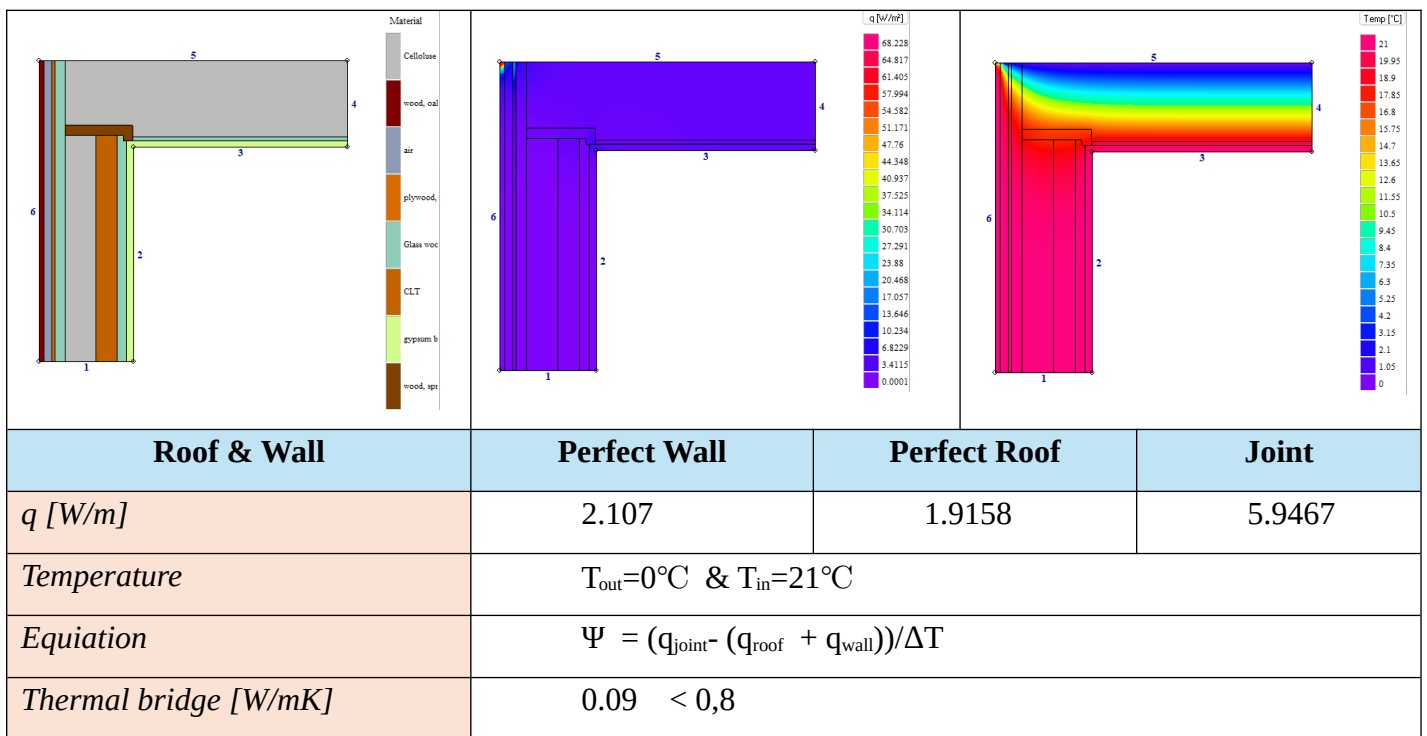


Figure 5.4.1.1: Wall To Roof Joint & Thermal heat loss calculation

5.4.2 Thermal Bridge Calculation for Connection Between Wall and Intermediate Slab

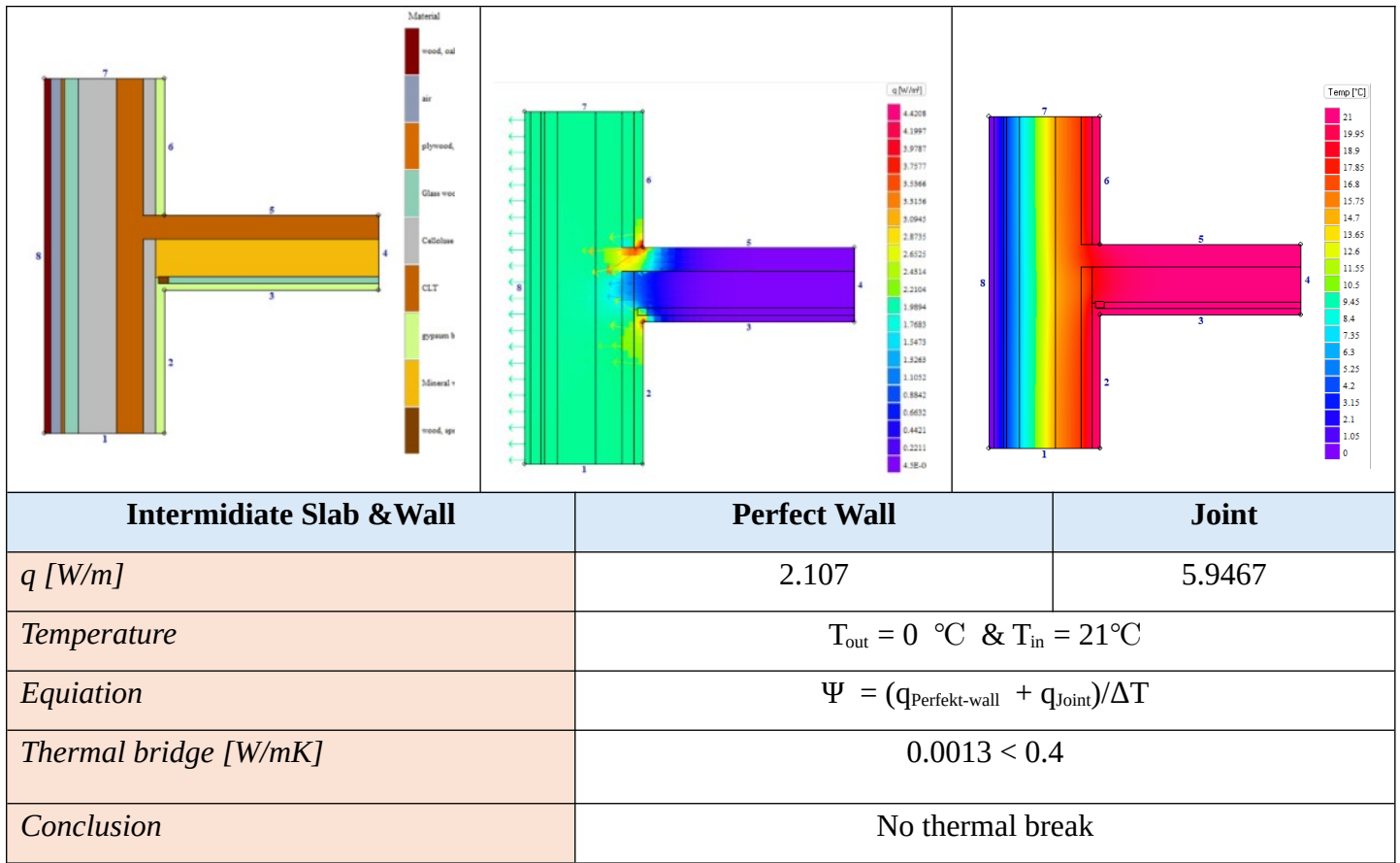
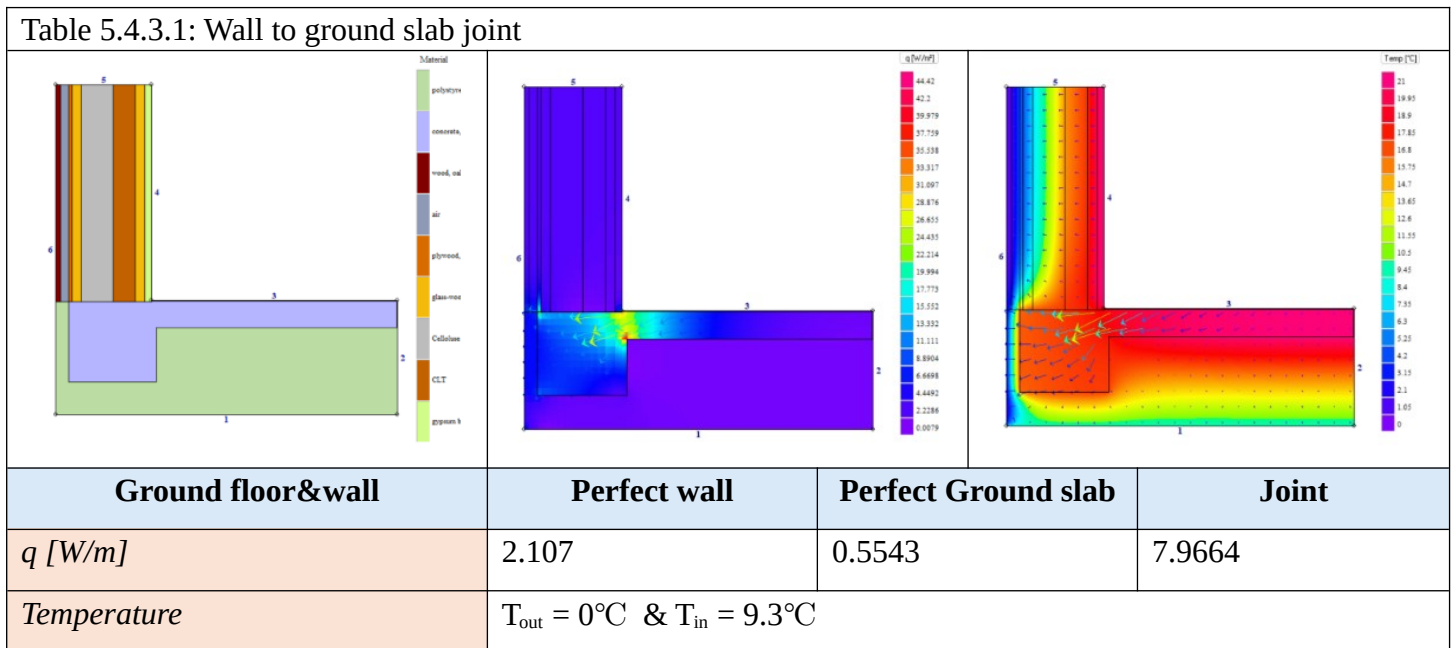


Figure 5.4.2.1: Wall to Intermediate slab joint & Thermal heat loss calculation

5.4.3 Thermal Bridge Calculation for Connection Between Wall and Ground Slab

Figure 5.4.3.1: Wall to ground slab joint & Thermal heat loss calculation



Equation	$\Psi = (q_{\text{joint}} - (q_{\text{ground}} + q_{\text{wall}})) / \Delta T$
Thermal bridge [W/mK]	0.25 < 0,8
Conclusion	No thermal break

5.5 Foundation Calculation

To ensure the foundation design meets moisture safety requirements, two key values for the ground slab were evaluated. The first was the temperature difference between indoor and outdoor environments, referred to as Delta T (ΔT). The second was the relative humidity (RH) beneath the concrete slab. The following calculations outline each step leading to this result.

The thermal conductivity of the clay is assumed to be 1,5 W/m°C

$$(1) L/B = 22/17 = 1,3$$

$$(2) d = \frac{d_i \cdot \lambda}{\lambda_i} = \frac{0,4 \cdot 1,5}{0,025} = 24$$

$$(3) d/B = 24/17 = 1,4$$

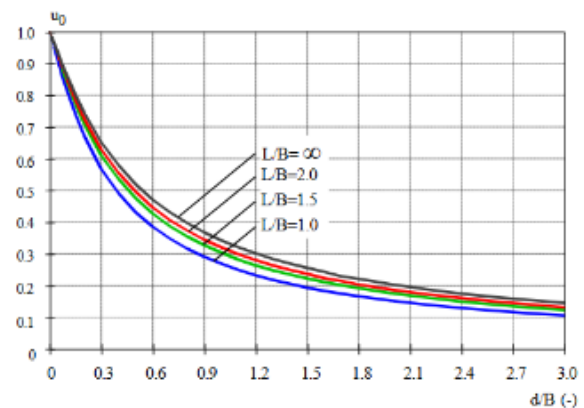
$$u_o = \frac{T_{\text{soil}} - T_e}{T_{\text{in}} - T_e} = 0,24 \rightarrow \text{from the figure we can estimate } u_o$$

$$\Delta T = (1 - u_o) (T_{\text{in}} - T_e)$$

$$\Delta T = (1 - 0,24) (21 - 9,3)$$

$$\Delta T = 8,892 \text{ } ^\circ\text{C}$$

In this case, ΔT is higher than the required value (3 °C). It is not necessary to add a vapour barrier to the central part of the foundation.



To calculate the relative humidity (RH) under the slab, data such as thermal conductivity, thermal resistance, layer temperatures, vapor permeability, saturation, and vapor resistance were collected. Some values were calculated, and others were taken from WUFI, as shown in Table 1 in the Appendix.

The results show that the RH is well below the critical level for mold growth. Under the concrete slab, the RH was 47.5%, far below the level needed for mold to form.

6. Discussion

6.1 Energy Performance & Thermal Comfort

The heating demand of both intermediate and final design achieves the requirement of the design goal, which are under 35kWh/m² per year. Total energy consumption is total energy consumption of 32832kWh, which is 13.6 kWh/m². It is a low heating demand, indicating that this residential building performs well considering energy efficiency.

The energy generated by PV panel is 29010kWh annually, which is smaller than the energy input that is needed for this building, however, the building achieves an energy balance of 88%. This near net-zero energy performance underscores the sustainability of the design.

Compared to initial design, A significant reduction in heating load indicates a clear improvement in energy efficiency, which is decrease by 52.3%. This also suggests enhanced building envelope performance and efficient mechanical systems.

The energy-efficient of building design is highly successful. The results demonstrate that sustainable building operations can be achieved through optimized solar energy utilization and well-designed insulation structures. Although there remains a 12% energy gap from achieving zero energy consumption, further optimizations can be implemented to achieve net-zero energy in the future.

In terms of indoor thermal comfort, it was a significant challenge in the initial design, with the building's overheating rate reaching 17%, a highly uncomfortable value well above acceptable standards. In contrast, the new design achieves an overheating proportion of only 2.4%, with certain areas experiencing no overheating at all, i.e., an overheating rate of 0.

This result demonstrates the new design is excellent and rational performance in preventing excessive solar radiation from entering the building. It effectively balances the benefits of solar energy utilization while mitigating its adverse effects. Such shading strategies offer valuable insights for future design applications.

However, the results of this report should be regarded as a reference or an exploration of optimizing building performance. In practical applications, factors such as occupant movement and equipment aging, which are difficult to control, may lead to discrepancies between the actual energy consumption of the building and the simulation results. To achieve more realistic simulations and analyses, it is necessary to incorporate additional data inputs, allowing the simulation outcomes to better align with real-world conditions and enabling more reliable building performance evaluations.

6.2 Moisture Risk Assessment

The risk assessment should be considered mainly into 2parts probability (likelihood) and severity (impact), the level definition are presented in Table.6.2.1. The risk assessment matrix

assessment is demonstrated in Table.6.2.2.

Table.6.2.1 Definition of level for risk assessment

Level	Probability	Severity
Very low	Close to impossible to happen	/
Low	Less to appear this situation	Minor impact with limited damage, easily repairable.
Medium	Sometimes occurs, but not usually happen.	Significant impact, potentially leading to economic losses, requiring resource investment for repair
High	Often occur this risk.	Severe impact, causing significant structural damage or posing a threat to human health, difficult to repair, potentially requiring reconstruction.

Table.6.2.2 Risk assessment matrix

Risk	Probability	Severity	Risk describes	Risk management
Mold growth	Very Low	Low	Under a large mold index level, and the range of RH value over the critical threshold, mold growth risk appears. (Qiao et al., 2024)	Control the moisture level indoors, drying the material recently. (“Why Mold Grows – and How to Prevent It - IFMA Knowledge Library,” 2024)
Corrosion	Low	High	The combination of a high-humidity environment and surface contaminants can lead to material corrosion.	Clean the surface recently and applying protective anti-corrosion coatings to the material surface.
Rainwater Intrusion	Medium	Medium	Rainwater infiltrates through surfaces or cracks into the structure, increasing the moisture content inside materials and causing structural damage.	Regularly repair structural cracks, enhance waterproof layers or apply waterproof coatings, and promptly remove rainwater from surfaces.
Thermal bridge	Very low	Low	Improper insulation design can affect the heat transfer path, leading to energy loss.	Improving the design of insulation materials to reduce heat transfer and interruptions in the insulation layer.
Material degradation	Low	High	The functionality of materials is reduced due to various external or internal environmental influences, and in severe cases, this may lead to material damage.	Utilize more durable materials.
Condensation	High	Medium	Condensation forms on the surface or inside the material due to the temperature and humidity differences between the external and internal environments. In result, increase the moisture level inside construction.	Control indoor temperature and humidity. Increase air change.

6.3 Project Progress

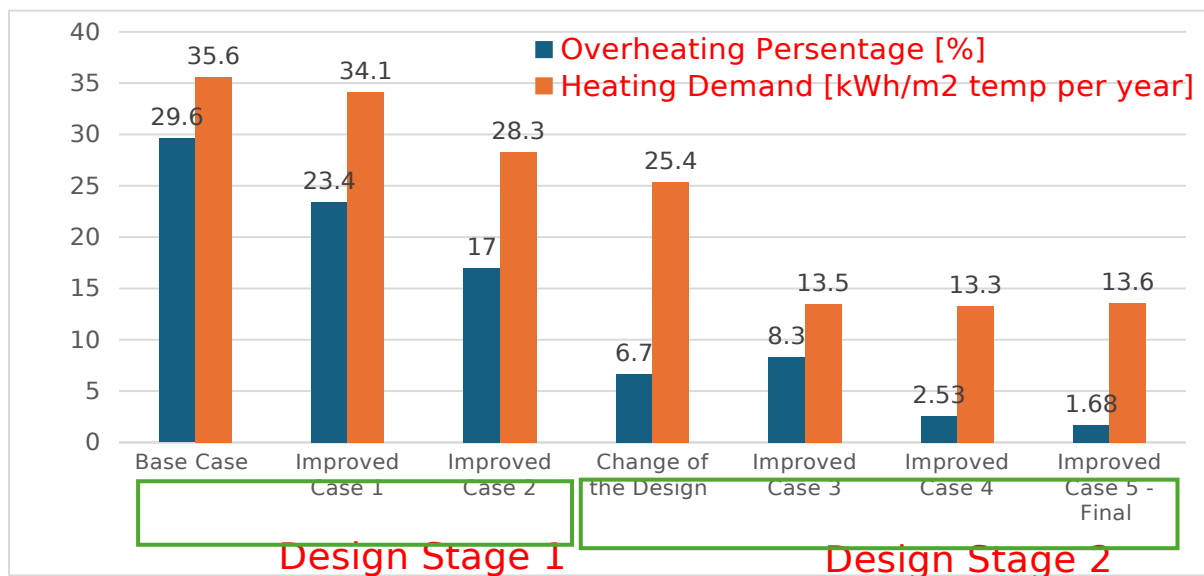


Figure.6.3.1 Project progress

The chart demonstrates the progression of Overheating Percentage and Heating Demand across iterative design stages, highlighting significant improvements in energy efficiency and thermal comfort.

In Design Stage 1, the Base Case shows an overheating percentage of 29.6% and a heating demand of 35.6 kWh/m²/year, which gradually improve in Improved Cases 2 and 3, reducing overheating to 17% and heating demand to 28.3 kWh/m²/year.

In Design Stage 2, a substantial redesign brings overheating down to 6.7% and heating demand to 25.4 kWh/m²/year. Further refinements in Improved Cases 4, 5, and 6 (Final) achieve minimal overheating (1.68%) and stabilize heating demand around 13.6 kWh/m²/year.

This iterative process highlights the effectiveness of design optimization in achieving both thermal comfort and energy efficiency, making the final design well-suited for sustainable and comfortable living.

7. Conclusion

This report presents the design and optimization process of a multi-family apartment building located in the Lund region. The report is divided into two main phases. In the first phase, the design met the requirements for energy consumption.

However, it is not fully complied with thermal comfort standards, and certain aspects were overlooked in the moisture study of the first design.

The final optimized design addressed the issues identified in the first phase, achieving nearly zero energy consumption standards, reducing potential structural moisture risks, and resolving overheating problems within the apartments.

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Appendix

Table 1: Calculation of Relative Humidity under the concrete slab.

Material /Position	Depth	Heat Cond	Heat Resistance	Temp	Diff In T	Vapor Saturation	Vapor Permeability	Vapor Resistance	Vapor Content	Diff In V	Relative Humidity	Assumed Vapor Content
	D (M)	Λ (W/M°C)	R (M²K/W)	T(°C)	Δt (°C)	V_s (G/M³)	Δ (M²/S)	Z (S/M)	V (G/M³)	Δv (G/M³)	Rh (%)	Out
Indoors, Rsi				21		18.32			8.99		0.49	In, Add On
			0.13		0.07			0		0		
				20.92		18.21			8.99		0.49	
Weather Resistive Barrier	0.001	2.3	0.0004		0.0002		0.0	1000		0.02		
				20.92		18.21			8.96		0.49	
Reinforced Concrete	0.125	1.6	0.078125		0.04		0.000009	13888.88		0.36		
				20.88		18.11			8.59		0.47	47,5 %
XPS	0.4	0.025	16		9.13		0.000008	50000		1.30		
				11.74		12.22			7.29		0.59	
Gravel	3	0.7	4.28		2.44		0.0000139	215827.33		5.65		
				9.3		8.27			2		0.24	
Outside, Rse				9.3		7.06			2		0.28	
			20.49	11.7				266827.33		6.99		